NRF announces eight inaugural projects under the AI-for-Science (AI4S) Initiative to accelerate scientific discovery

[SINGAPORE, 16 JUNE 2026] – The National Research Foundation, Singapore (NRF) has **launched eight research projects under the AI-for-Science (AI4S) Initiative**, a S$120 million national programme aimed at harnessing Artificial Intelligence (AI) to accelerate scientific discovery across thematic areas of interest to Singapore. The Initiative brings together teams of AI researchers and scientific domain experts from Singapore and top international institutions to develop AI methods and tools that can raise research productivity and accelerate discovery.

Professor Tan Chorh Chuan, Permanent Secretary (National Research and Development), announced the official start of the eight inaugural projects at the AI4X-Accelerate Conference 2026. Recognising the potential for AI to transform and accelerate research and scientific discovery, the AI4S Initiative supports the National AI Strategy 2.0 (NAIS 2.0), which aims to build a vibrant and innovative AI research ecosystem.

Harnessing AI to accelerate scientific discovery

Scientific discovery has traditionally been constrained by the scale of what researchers can observe, test and analyse. In recent years, we have seen the disruptive impact of AI on how and how fast scientific discovery can be done. AI is now increasingly being used to automate labour-intensive tasks (e.g. literature reviews and data analyses), explore far more hypotheses than previously possible, and uncover patterns in vast datasets that would otherwise have been difficult or impossible for researchers to have done.

The AI4S Initiative is designed to harness the disruptive potential of AI in a sustained and systematic way. One key enabler is the co-location of research teams on the CREATE campus to promote deep collaborations between researchers from different disciplines, particularly AI and computational experts and researchers with deep expertise in scientific domains.

The Initiative serves also to support knowledge transfer and capability-building within Singapore's research community. It will cultivate a new generation of "bilingual" scientists, i.e. researchers who are proficient in both AI and the sciences. These scientists will play a crucial role in bridging AI and scientific disciplines, interpreting real-world data and translating complex domain needs into precise computational models and algorithms to accelerate scientific discovery.

The Initiative entails two grant schemes. The **Challenge grant** supports larger-scale projects aimed at advancing the development of AI methods and tools for specific use areas, such as the eight inaugural projects announced. The **Catalytic grant** supports smaller-scale exploratory projects using AI methods and tools for frontier science. The eight projects selected under the Challenge grants cover multiple thematic areas of interest to Singapore. These areas include advanced manufacturing & materials, biomedical & health sciences, energy & climate resilience, digital trust, and aviation & maritime technologies.

Across both grant schemes, the AI4S Initiative is expected to support up to 20 teams from leading research institutions globally across five countries, namely Singapore, the United

Kingdom, the United States, Japan and Canada. The participating institutions include the National University of Singapore (NUS), Nanyang Technological University (NTU), the Agency for Science, Technology and Research (A*STAR), Imperial College London, the University of Cambridge, the University of Toronto, the University of Illinois Urbana-Champaign and Kyoto University. Please see the Annex for more details.

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About the AI-for-Science (AI4S) Initiative

AI4S is a national funding initiative by Singapore's National Research Foundation (NRF). It aims to advance the application of artificial intelligence to transform the pace and scope of scientific discovery. The Initiative brings together collaborative teams of AI and domain researchers from leading institutions in Singapore and internationally, nurturing bilingual scientists fluent in both AI and their respective scientific fields. Through its Challenge and Catalytic grant schemes, AI4S supports both ambitious larger-scale research projects and creative, exploratory ideas at the frontier of science, in domains such as life sciences, materials science and quantum science.

About the National Research Foundation, Singapore (NRF)

The National Research Foundation, Singapore (NRF) is a department within the Prime Minister's Office. It sets the national direction for research and development (R&D) by developing policies, plans and strategies for research, innovation and enterprise. It also funds strategic initiatives and builds up R&D capabilities by nurturing research talent. Learn more about NRF at https://www.nrf.gov.sg

ANNEX

Inaugural AI4S Projects – Summary

No.	Project	Host Institution
1	Revolutionising Energy Transport: AI-Engineered Interfaces for Advanced Energy Carrier Dynamics	NTU
2	Accelerating Genomic Research with Artificial Intelligence: From Data to Discovery	NUS
3	AI for Program Reasoning	Imperial Global Singapore
4	Materials Data Foundry: Accelerating Synthesis of Complex Materials for Future Applications	NUS
5	Generative AI Models for the Creation and Delivery of mRNA Vaccine	A*STAR Research Entities (ARES)
6	KGAI4Ag: Advancing Knowledge-Guided AI to Develop Agricultural Digital Twins for Singapore's Climate Resilience	Illinois Advanced Research Center at Singapore Ltd. (Illinois ARCS)
7	Overcoming the Terascale Design Challenge: Next-Generation Efficient AI for Discoveries in Surface Science and Catalysis	A*STAR Research Entities (ARES)
8	BloodCounts! – Artificial Intelligence for the Full Blood Count Test	Cambridge Centre for Advanced Research and Education in Singapore Ltd. (CARES)

Inaugural AI4S Projects – Details

Project 1: Revolutionising Energy Transport: AI-Engineered Interfaces for Advanced Energy Carrier Dynamics
Co-led by NTU and Kyoto University

Modern energy devices, from batteries that store electricity to electrolysers that produce hydrogen, depend not just on the materials inside them, but on how different materials interact at their surfaces and boundaries. These interaction zones, called interfaces, are poorly understood and have long been a bottleneck in building better energy systems.

This project develops an AI-powered research platform designed to discover better interfacial materials faster, known as "MOITAI" (MOF-based Optimised Interface with Artificial Intelligence). At its core are Metal-Organic Frameworks (MOFs), a class of nano-scale materials with highly tuneable, porous structures that can be engineered to control how energy carriers such as protons, ions, electrons and molecules move across material boundaries.

MOITAI combines robotics-assisted experiments, real-time material analysis, and multiple forms of AI, including generative AI and physics-informed neural networks (AI models that incorporate the laws of physics to make more reliable predictions), in a continuous feedback loop. This allows the team to rapidly test and refine new material combinations.

The platform targets three practical goals: improving ion transport for flow batteries (a type of large-scale rechargeable battery), designing electrically conductive interfaces for devices like smart windows that change colour on demand, and developing chemical sensors with the sensitivity to detect explosive or flammable molecules at trace concentrations. The research is supported by Mitsubishi and benefits from access to Japan's most advanced synchrotron facilities (large-scale machines that use powerful X-rays to study material structures in real time).

Project 2: Accelerating Genomic Research with Artificial Intelligence: From Data to Discovery
*Co-led by NUS and A*STAR Research Entities (ARES)*

The volume of human genetic data being generated worldwide is growing faster than scientists can meaningfully analyse. Large-scale initiatives such as Singapore's own SG100K programme, which is building a genomic database of 100,000 Singaporeans, are producing rich datasets, but existing AI tools tend to examine DNA, RNA and gene activity in isolation, missing the connections between them that are often most medically significant.

This project develops MultiOmicsFM, a unified AI foundation model (a large, general-purpose AI system trained on vast datasets and adaptable to many tasks) designed to read and interpret DNA sequences, RNA sequences and gene expression profiles together. Think of it as moving from separate diagnostic tests to an integrated health report that captures how genes are structured, regulated and expressed in concert.

The project builds out in four stages: (i) curating high-quality genomic datasets from Singapore and global sources; (ii) designing a scalable AI architecture optimised for biological data; (iii) developing AI agent tools that help biologists work alongside AI to accelerate discovery; and (iv) validating the model in real-world applications such as predicting disease risk, interpreting genetic variants associated with illness, and optimising mRNA-based therapies.

By harnessing Singapore's uniquely multi-ethnic genomic datasets, the project aims to position Singapore as a global leader in AI-driven precision medicine, i.e. the practice of tailoring medical treatment to an individual's genetic profile.

Project 3: AI for Program Reasoning
Co-led by Imperial Global Singapore and NUS

Software bugs are costly and, in critical systems, potentially dangerous. As AI-generated code becomes more common, reducing the time it takes to write software, the need to verify that the code actually works as intended becomes more urgent, not less.

This project builds AI tools capable of automatically reasoning about computer programs: checking that they behave correctly, inferring what they are supposed to do, and, in the most rigorous cases, constructing formal mathematical proofs of correctness. The researchers are tackling both formal reasoning (proving, with mathematical precision, that a programme meets a specification) and informal reasoning (learning the intended behaviour of software that was never fully documented, which is the reality in most real-world code-bases).

The practical targets are demanding: automated analysis and verification of complex network protocol implementations (the rules governing how computers communicate), consensus protocol implementations (the algorithms that allow distributed systems like blockchains to agree on shared data), and components of the Linux operating system kernel (the foundational software layer that runs on billions of devices).

The result will be specialised (agentic) AI-powered tools that can help developers and security professionals catch errors in critical software, and lay the groundwork for reliably auditing the large volumes of AI-generated code that are likely to become standard in software development. The vision is thus towards AI-based Verification and Validation (V & V) of AI-generated code.

Project 4: Materials Data Foundry: Accelerating Synthesis of Complex Materials for Future Applications
Co-led by NUS and University of Toronto

Developing new materials, including those used in more efficient computer chips, longer-lasting batteries, and low-cost equipment for producing green hydrogen, typically takes years of slow, trial-and-error laboratory work. AI has shown promise in speeding this up, but most current AI models work with idealised theoretical structures and ignore real-world complexities such as how materials are actually made, how defects form, or how materials behave at different scales.

This project establishes the Materials Data Foundry (MDF), a purpose-built facility that combines high-throughput robotic laboratories, real-time in-situ characterisation (the ability to observe materials as they are being formed), and advanced AI. The MDF is designed to generate what would be the world's most comprehensive experimental dataset of real materials and their synthesis routes, data that current AI models simply lack.

The AI trained on this dataset will be capable of predicting not only what a material's properties will be, but also how to actually make it, closing a critical gap between theoretical design and practical production. Target application areas include energy (materials for hydrogen production and storage), electronics (next-generation semiconductors), and infrastructure (corrosion-resistant alloys).

Project 5: Generative AI Models for the Creation and Delivery of mRNA Vaccine
*Led by A*STAR Research Entities (ARES)*

mRNA vaccines, brought to global attention during the COVID-19 pandemic, work by delivering genetic instructions to cells, prompting the body to produce proteins that train the immune system. They can be designed and manufactured relatively quickly, making them promising solutions not just for infectious diseases but also for personalised cancer treatment. However, designing the right mRNA molecule, and ensuring it reaches the right cells in the body reliably and safely, remain significant technical challenges.

This project builds an AI agent system (a set of coordinating AI models that work together towards a shared goal) to automate and optimise the entire mRNA vaccine design process. This includes engineering the mRNA sequence itself (both its protein-coding section and the structural regions that influence how stable and effective it is), as well as the lipid nanoparticles (LNPs) used to deliver it. LNPs are microscopic fat-based capsules that protect the mRNA and help it enter target cells; their precise composition affects where in the body the vaccine is delivered and how strong an immune response it triggers.

The system integrates multiple AI components: a foundation model trained on protein and RNA sequences; a retrieval-augmented generation (RAG) framework that draws on curated scientific literature and data to inform design decisions; deep learning models to assess mRNA stability; and a multi-modal AI model to optimise delivery. Together, these are coordinated by an overarching AI agent that iteratively refines designs based on experimental feedback. The goal is to reduce the time needed to design an optimised mRNA vaccine candidate from months to hours.

Project 6: KGAI4Ag: Advancing Knowledge-Guided AI to Develop Agricultural Digital Twins for Singapore's Climate Resilience
Co-led by Illinois Advanced Research Center at Singapore Ltd. (Illinois ARCS) and NUS

Southeast Asia's food systems face growing pressures from climate change. Increasingly erratic rainfall, rising temperatures and shifting growing seasons are threatening staple crops such as rice and key export commodities like oil palm, cacao and coffee, with knock-on effects for food security and trade across the region.

This project builds digital twins for ASEAN agro-ecosystems, which are virtual replicas of real farmland that are continuously updated with data from satellites and ground-level sensors, and can simulate how different conditions or decisions play out before they are implemented. The twins are powered by Knowledge-Guided AI (KGAI): AI models that are designed not just to find patterns in data but to respect and incorporate established scientific understanding of how crops grow and how ecosystems function. This makes them more interpretable and reliable than pure data-driven approaches.

Practical tools developed from the platform will include AI-driven forecasting for food trade flows and decision-support systems that help farmers and policymakers optimise planting strategies, resource use, regional policy design and supply chain logistics in response to changing conditions.

The project aims to position Singapore as a regional hub for AI-driven agricultural innovation, contributing tools and frameworks that can be adapted by other ASEAN countries facing similar food security challenges.

Project 7: Overcoming the Terascale Design Challenge: Next-Generation Efficient AI for Discoveries in Surface Science and Catalysis
*Co-led by A*STAR Research Entities (ARES) and Imperial Global Singapore*

Catalysts are materials that speed up chemical reactions without being consumed in the process. They are essential to producing clean fuels, reducing industrial emissions and making high-value chemicals. The challenge is that the number of possible catalyst combinations, across different elements, structures and surface arrangements, is astronomically large, making systematic discovery by conventional laboratory methods prohibitively slow and expensive.

This project develops a Surface Science Foundation Model: a large AI model trained on an estimated two million or more computational and experimental data points describing how atoms behave on material surfaces. By learning from this vast dataset, the model can predict which candidate materials are likely to be effective catalysts, and guide the automated design of new ones, a process known as inverse design, where researchers start from the desired outcome and work backwards to identify what material would achieve it.

The AI framework incorporates a method called Equivariant Sparse Structure Learning (ESSL), state-of-the-art equivariant learning architectures and sparse learning techniques, which enables accurate predictions with significantly less computing power, making the approach practical at scale. An agentic optimisation workflow then coordinates multiple AI techniques, including reinforcement learning and Bayesian optimisation, to refine candidate materials against multiple performance criteria simultaneously.

Target applications include sustainable chemical production pathways, such as converting carbon dioxide into fuels (a pathway for carbon utilisation) and valorising glycerol, a by-product of biodiesel production, into high-value specialty chemicals. The project aims to reduce the time and cost of identifying promising catalyst candidates by at least five times.

Project 8: BloodCounts! – Artificial Intelligence for the Full Blood Count Test
Co-led by Cambridge Centre for Advanced Research and Education in Singapore Ltd. (CARES) and NTU

The full blood count (FBC) is one of the most routinely ordered medical tests in the world. It measures features such as haemoglobin levels, the number of red and white blood cells, and platelet counts, and is used to screen for a wide range of conditions from anaemia to infection. The machines that run FBC tests actually generate far richer raw data than the summary report that doctors receive, but this underlying data is almost always discarded.

Research at Cambridge has shown that this discarded raw data contains clinically valuable signals: it can help detect iron deficiency, identify early-stage kidney cancer, distinguish between different forms of leukaemia (blood cancer), and assess how well platelets function in forming blood clots. At SingHealth, around 20% of FBC tests are supplemented with a blood smear, a microscopic image of blood cells examined by trained staff, adding a further source of visual information that can support diagnosis.

This project develops a Foundation Model of Blood: a large, general-purpose AI system trained on data from millions of FBC samples that combines raw numerical data and blood cell imaging to detect patterns associated with disease. It will be trained in an agnostic manner, meaning it is not optimised for any single condition, making it adaptable to a wide range of clinical uses. Initial applications will focus on stroke and lung cancer studies.

The model will be open-sourced for use by researchers and clinicians in Singapore and globally, making it particularly valuable in rare disease contexts where any single hospital’s sample numbers are too small to train reliable AI tools independently.